In [91]:
```
import plotly.express as px
import plotly.graph_objects as go
import plotly.io as pio
import plotly.colors as colors
pio.templates.default = "plotly_white"
import pandas as pd
```

In [92]:
```
data = pd.read_csv("Sample - Superstore.csv",encoding = "latin-1")
data.head()
```

Out[92]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	
0	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Hend
1	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Hend
2	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	An
3	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Laude
4	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Laude

5 rows × 21 columns

Let's start by looking at the descriptive statistics of the dataset

In [93]:
```
data.describe()
```

Out[93]:

	Row ID	Postal Code	Sales	Quantity	Discount	Profit
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203	28.656896
std	2885.163629	32063.693350	623.245101	2.225110	0.206452	234.260108
min	1.000000	1040.000000	0.444000	1.000000	0.000000	-6599.978000
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000	1.728750
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000	8.666500
75%	7495.750000	90008.000000	209.940000	5.000000	0.200000	29.364000
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000	8399.976000

In [94]:

```
data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   str
 2   Order Date     9994 non-null   str
 3   Ship Date      9994 non-null   str
 4   Ship Mode      9994 non-null   str
 5   Customer ID    9994 non-null   str
 6   Customer Name  9994 non-null   str
 7   Segment        9994 non-null   str
 8   Country        9994 non-null   str
 9   City           9994 non-null   str
 10  State          9994 non-null   str
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   str
 13  Product ID     9994 non-null   str
 14  Category       9994 non-null   str
 15  Sub-Category   9994 non-null   str
 16  Product Name   9994 non-null   str
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(3), int64(3), str(15)
memory usage: 1.6 MB
```

Converting Date Columns

In [95]:

```
data['Order Date'] = pd.to_datetime(data['Order Date'])
data['Ship Date'] = pd.to_datetime(data['Ship Date'])
```

Adding New Date Based Column

In [96]:

```
data['Order Month'] = data['Order Date'].dt.month
data['Order Year'] = data['Order Date'].dt.year
data['Day of week'] = data['Order Date'].dt.dayofweek
```

In [97]:

```
data.head()
```

Out[97]:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	...
0	1	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...
1	2	CA-2016-152156	2016-11-08	2016-11-11	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	...
2	3	CA-2016-138688	2016-06-12	2016-06-16	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	...
3	4	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...
4	5	US-2015-108966	2015-10-11	2015-10-18	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	...

5 rows × 24 columns

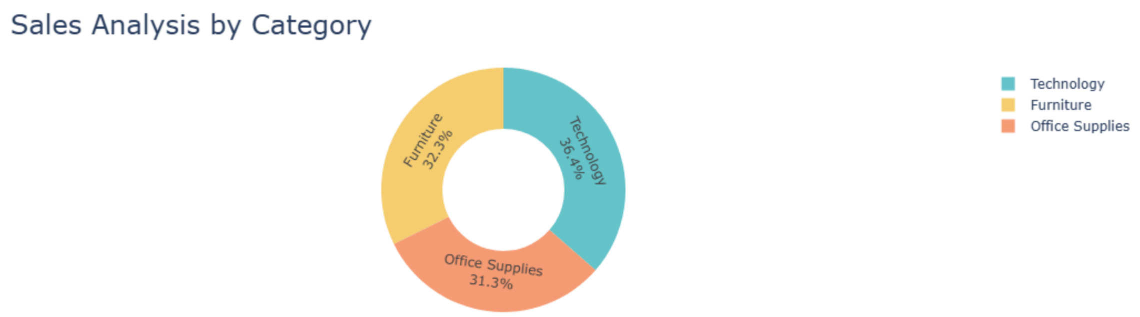
Q.1 Monthly Sales Analysis

In [98]:
```
Sales_by_Month = data.groupby('Order Month')['Sales'].sum().reset_index()
fig = px.line(Sales_by_Month,
             x="Order Month",
             y='Sales',
             title ='Monthly Sales Analysis')
fig.show()
```

Q.2 Sales Analysis By Category

In [99]:
```
Sales_by_category = data.groupby ('Category')['Sales'].sum().reset_index()
```

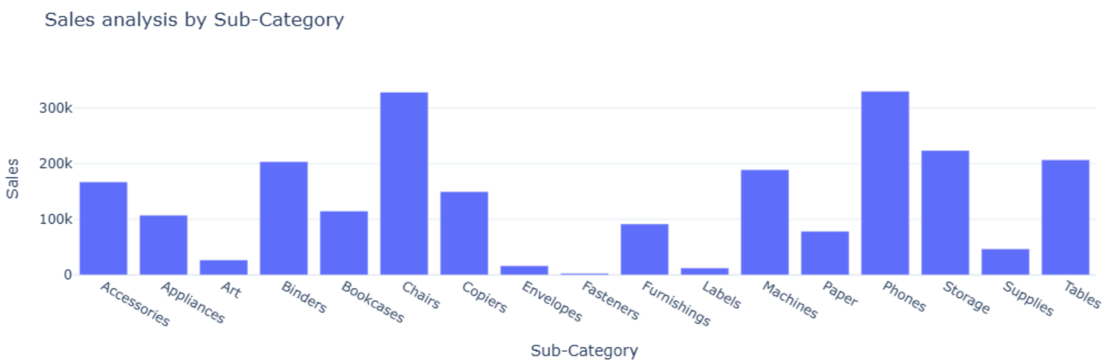
```python
fig = px.pie(Sales_by_category,
             values = 'Sales',
             names = 'Category',
             hole = 0.5,
             color_discrete_sequence = px.colors.qualitative.Pastel)

fig.update_traces(textposition='inside',textinfo='percent+label')
fig.update_layout(title_text = 'Sales Analysis by Category',title_font=dict(size = 24))
fig.show()
```



Q.3 Sales Analysis By Sub-Category

In [100…

```python
sales_by_Subcategory = data.groupby('Sub-Category')['Sales'].sum().reset_index()

fig = px.bar(sales_by_Subcategory , x = 'Sub-Category', y = 'Sales',title='Sales analysis b
fig.show()
```



Q.4 Monthly Profit Analysis

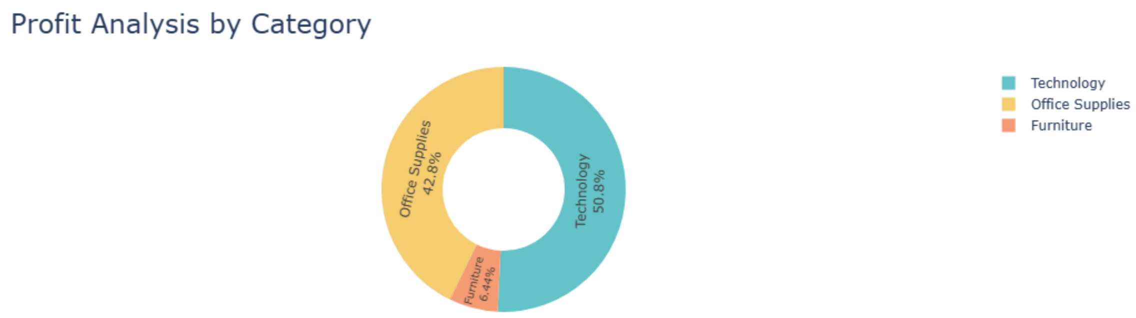
In [101…

```python
profit_by_month = data.groupby('Order Month')['Profit'].sum().reset_index()
fig = px.bar(profit_by_month,
             x='Order Month',
             y= 'Profit',
             color_discrete_sequence=['orange'],
             title = 'Monthly Profit Analysis')
fig.show()
```

Q.5 Profit Analysis By Category

In [102…

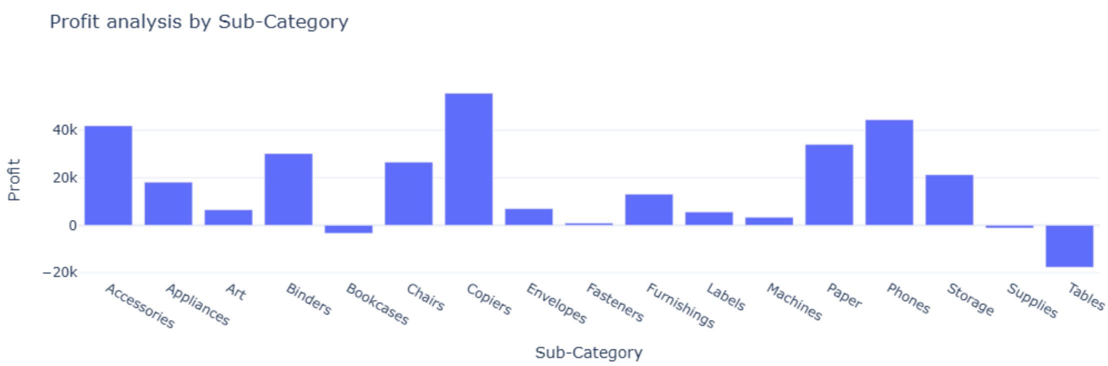
```
profit_by_category = data.groupby('Category')['Profit'].sum().reset_index()
fig = px.pie(profit_by_category,
             values = 'Profit',
             names = 'Category',
             hole = 0.5,
             color_discrete_sequence = px.colors.qualitative.Pastel)

fig.update_traces(textposition='inside',textinfo='percent+label')
fig.update_layout(title_text = 'Profit Analysis by Category',title_font=dict(size = 24))
fig.show()
```



Q.6 Profit Analysis By Sub-Category

In [103…

```
profit_by_subcategory = data.groupby('Sub-Category')['Profit'].sum().reset_index()
fig = px.bar(profit_by_Subcategory , x = 'Sub-Category', y = 'Profit',title='Profit analysi
fig.show()
```



Q.7 Sales and Profit Analysis By Customer Segment

In [104…

```
sales_profit_by_segment = data.groupby('Segment').agg({'Sales': 'sum','Profit':'sum'}).rese

color_palette = colors.qualitative.Pastel

fig = go.Figure()
fig.add_trace(go.Bar(x=sales_profit_by_segment['Segment'],
                     y=sales_profit_by_segment['Sales'],
                     name='Sales',
                     marker_color=color_palette[0]))

fig.add_trace(go.Bar(x=sales_profit_by_segment['Segment'],
                     y=sales_profit_by_segment['Profit'],
                     name='Profit',
                     marker_color=color_palette[1]))

fig.update_layout(title='Sales and Profit Analysis by Customer Segment',
                  xaxis_title='Customer Segment', yaxis_title='Amount')

fig.show()
```

Q.8 Analyse Sales - To - Profit Ratio

In [105…

```
sales_profit_by_segment = data.groupby('Segment').agg({'Sales': 'sum', 'Profit': 'sum'}).re
sales_profit_by_segment['Sales_to_Profit_Ratio'] = sales_profit_by_segment['Sales'] / sales
print(sales_profit_by_segment[['Segment', 'Sales_to_Profit_Ratio']])
```

```
       Segment  Sales_to_Profit_Ratio
0     Consumer               8.659471
1    Corporate               7.677245
2  Home Office               7.125416
```

In [ ]: